

Design, Fabrication, and Testing of a Silicon Photodiode

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Abstract

This paper presents the design, fabrication, simulation, and characterization of a silicon p-n photodiode. The fabrication process included wafer cleaning, thermal oxidation, photolithography, reactive ion etching, ion implantation, annealing, and back-end-of-line contact formation. This project connects photodiode physics, such as depletion region formation, photocurrent generation, dark current, and responsivity, to the actual cleanroom steps used to build the device. Synopsys Sentaurus TCAD was also used to simulate the fabrication process and predict how process parameters affect the final device structure and performance. The simulations showed that implantation energy affects the dopant depth, peak dopant concentration, and junction depth, which can influence leakage current and optical response. During fabrication, oxide over-etching and incomplete passivation affected the final device yield. Electrical and optical characterization of available devices showed non-ideal behavior, including low responsivity, unstable photocurrent, and possible contact-related issues. Overall, the results show how small changes in lithography, etching, passivation, and contact formation can strongly affect photodiode performance.

Index Terms— Silicon photodiode, semiconductor fabrication, photolithography, reactive ion etching, ion implantation, thermal oxidation, Sentaurus TCAD, device characterization.

Introduction

Semiconductor and Device Physics Background

Semiconductor devices are used in nearly every electronic appliance in use today. These tiny devices control how electrical signals move through a circuit, making them important in computers, phones, sensors, communication systems, and many other technologies.

In semiconductors, the conductivity of a material is increased by introducing donor impurities into the silicon in a process called doping. The donor dopants, such as phosphorus or arsenic, introduce additional free electrons into the silicon lattice when replacing silicon atoms, creating an n-type material. Additionally, acceptor dopants, such as boron, create holes, producing a p-type material. The arrangement of p-type and n-type areas creates the PN junction needed for photodiodes [1].

A photodiode is a semiconductor device that transforms light into an electrical signal utilizing the photoelectric effect [2]. When photons emitted from a light source hit the photodiode, it creates electron-hole pairs that move through the device, creating electric current.

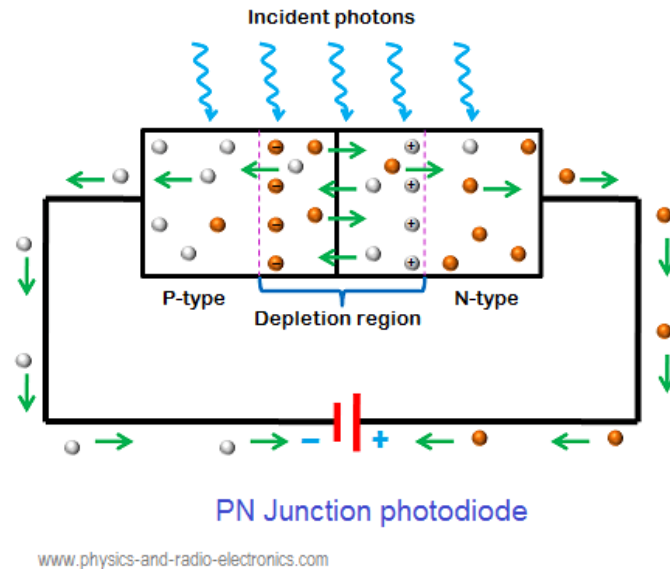


Figure 1: Diagram of PN circuit. [3]

A PN junction is at the heart of a photodiode device, creating three distinct sections that make photon detection possible. From Figure 1, we can see the P-type region, the N-type region, and the depletion region. The P-type region contains charge carriers called holes, while the N-type region contains electrons. The central area, called the depletion region, contains immobile ions, creating an electric field that pulls charges apart, causing them to move by drift. By separating the charge carriers before they can recombine, this allows the photodiode to generate photocurrent. Charge carriers that are generated outside the depletion region can also generate photocurrent by diffusing into the depletion

region, where they will also be separated by the electric field [1]. Charge carriers can move either by drifting or diffusion. Drifting is when carriers are moved quickly across the junction by the electric field. Outside the depletion region, drift occurs when the charge carriers have to slowly spread toward the junction, which takes more time. If that diffusion distance is large, it can slow down the overall response of the device.

When the p-type and n-type regions are in contact, the electrons and holes will naturally move across the boundary due to the carriers on the other side. As these carriers recombine, they leave behind stationary fixed charged atoms, creating an electric field and voltage across the junction. This then bends the energy bands and forms the depletion region [4].

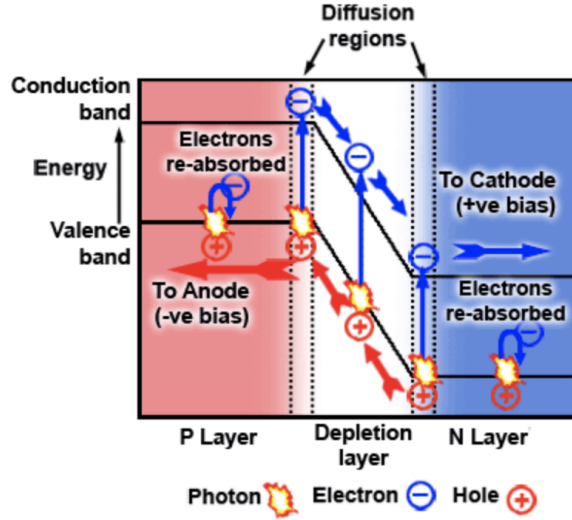


Figure 2: Energy Band Diagram of Photodiode Action. [5]

Figure 2 shows the energy band diagram for a PN junction in reverse bias. The difference in doping between the p-type and n-type regions result in a band bending, creating the electric field in the depletion region. The separation and the collection of the charge carriers in the device produces a photocurrent that can flow through the circuit.

Additionally, the circuit system is reverse biased, meaning that the P-type region is connected to the negative terminal and the N-type region is connected to the positive terminal. By reverse biasing the photodiode, the holes and electrons are drawn to the edges of their respective regions. Reverse biasing increases the width of the depletion region and strengthens the electric field. This makes it easier to separate and collect charge carriers, and also allows the photodiode to respond faster to incoming light. This also reduces junction capacitance, leading to increased speed and bandwidth.

Photodetection can occur when the following condition is satisfied [4]. See Equation 1.

$$h\nu \geq E_g \quad (1)$$

In Equation 1, h is Plank's constant, ν is the photon frequency, and E_g is the bandgap energy. When the photons are able to meet the energy required, they are absorbed into the device, creating electron-hole pairs. The depth of absorption will depend on the wavelength of the photon and the material's optical qualities.

The efficiency of converting photons into electrical current can be characterized by responsivity.

$$R = \eta \frac{q}{h\nu} \quad (2)$$

In Equation 2, η is the quantum efficiency, $h\nu$ is the energy of the photon expressed using Planks constant and the frequency of light, and q is the charge of a single electron. This equation shows that higher quantum efficiency leads to higher responsivity. This improves photodiode's ability to generate current from incident light.

The width of the depletion region depends on the doping levels in the semiconductor and the applied reverse-bias voltage, and can be described by Equation 3.

$$W = \sqrt{\frac{2\epsilon_s}{q} \left(\frac{N_A + N_D}{N_A N_D} \right) (V_{bi} + V_R)} \quad (3)$$

In Equation 3, W is the depletion region width, ϵ_s is the permittivity of silicon, q is the elementary charge, N_A and N_D are the acceptor and donor doping concentrations, V_{bi} is the built in potential, and V_R is the applied reverse bias voltage.

The depletion region also determines the junction capacitance, which is given in the following equation.

$$C_j = \frac{\epsilon_s A}{W} \quad (4)$$

In Equation 4, C_j is the junction capacitance, ϵ_s is the permittivity of the semiconductor, A is the junction area and W is the depletion region width. This shows that as the depletion region widens when under reverse bias, the junction capacitance will decrease. A lower capacitance will allow for faster charging and discharging of the device, leading to better response speed and bandwidth of the photodiode.

The response speed of a photodiode is sometimes limited by its RC time constant, which determines the bandwidth. This relation can be given by the following equation.

$$f_{3dB} = \frac{1}{2\pi RC} \quad (5)$$

In Equation 5, f_{3dB} is the cutoff frequency, R is the load resistance, and C is the junction capacitance. This relationship shows how reducing capacitance through increased depletion width improves high-speed performance.

Photodiode Operating Modes and I–V Characteristics

There are two primary operating modes in photodiodes that affect the performance and noise of the system. Firstly, photovoltaic mode is when a photodiode is operated with no external bias, relying on the built in electric field of the photodiode in order to generate a signal. This results in slow response times and low bandwidth, but offers low noise and minimal dark current. The best use for photovoltaic systems are in appliances such as light meters and solar cells.

The current–voltage behavior of a photodiode in dark current can be described by the Shockley diode equation.

$$I = I_s(e^{\frac{qV}{kT}} - 1) \quad (6)$$

In Equation 6, I_s is the saturation current, V is the applied voltage, q is the elementary charge, k is Boltzmann's constant, and T is temperature. This equation shows that even without illumination, a small current flows due to thermally generated carriers.

On the other hand, the photoconductive mode operates the photodiode in reverse bias. This both increases the electric field strength and widens the depletion region. This leads to a faster response time and higher bandwidth because it accelerates charge carrier separation and reduced the junction capacitance. However, when operating in photoconductive mode, higher dark current and increased noise must be accounted for. Due to its high speed capabilities, these devices have many uses in optical sensors, and fiber optic communications [2].

The different operating modes can be shown in the current-voltage characteristics. See Figure 3.

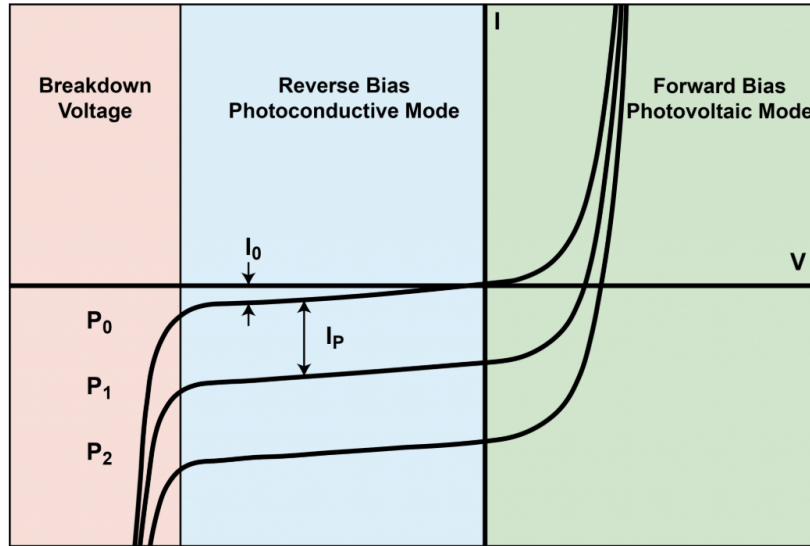


Figure 3: I-V Curve of Photodiodes. I_0 is Dark Current. I_P is photocurrent. P shows current at different light levels (P_0 is no incident light). [6]

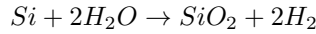
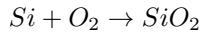
Motivation and Applications of Photodiodes

Learning and participating in the photodiode fabrication process is not only useful in connecting classroom theory to real-world application, but also introduces students to industry standard equipment, processes, and procedure. Fabricating photodiodes specifically is a preferred choice for this lab due to the fact that photodiodes only have a single p-n junction and minimal lithography and doping steps, while other semiconductor devices have multiple, precise, processes that often cannot be completed by available lab equipment.

This project allows students to observe p-n junction physics, depletion region formation, and electron-hole pair generation. This directly links the semiconductor fabrication processes to device behavior that can be measured.

Fabrication Process

Photodiode fabrication includes the following process that aid in a student's understanding of industry operations. Firstly, thermal oxidation is the process in which a layer of SiO_2 is grown on a silicon wafer in order to mask against implant or diffusion of dopant into silicon, surface passivation, device isolation, electrical isolation of multi-level metallization systems [1, 7]. The two primary methods for this process include dry and wet oxidation. The former using O_2 gas to create precise and thin oxides, while the latter uses water as an oxidant to create thick oxides at a faster rate. The chemical equations for these process are as follows.



The completion of the thermal oxidation process yields a clean $\text{Si} - \text{SiO}_2$ surface that provides the needed insulation and masking layer needed for future steps.

The growth of silicon dioxide during thermal oxidation can be described using the Deal-Grove model.

$$x^2 + Ax = B(t + \tau) \tag{7}$$

In Equation 7, x is the oxide thickness, t is the oxidation time, A is the linear rate constant, B is the parabolic rate constant, and τ is any initial oxide thickness. The Ax term describes reaction-limited growth at the silicon surface, which is important when the oxide layer is thin. The x^2 term represents diffusion limited growth through the oxide, which becomes important as the oxide gets thicker.

The constants A and B depend on factors like temperature and the oxidation environment, such as if it is dry or wet oxidation. With wet oxidation, oxide grows faster because diffusion is easier. The Deal-Grove model is important because it helps control oxide thickness, which ties into the masking process and junction depth.

Next, using photomasks and photolithography, desired patterns and designs can be copied onto a wafer. First, a photoresist material, S1827, is used to coat the polished chrome face of a silicon wafer. The wafer is then transferred a mask aligner with a selected photomask and exposed to UV light, this process makes the exposed regions more soluble, allowing it to be removed when rinsed with MF 319 developer and DI water.

The S1827 is a positive photoresist material, however negative photoresist can also be depending on the desired effect. The key difference between the positive and negative photoresist is their behavior when exposed to UV light. In a positive photoresist, the areas that are exposed will become dissolvable in water and easy to remove while in a negative photoresist, the areas exposed will become difficult to dissolve, making it harder to remove. For the S1827, the exposure to UV light will make the areas exposed more soluble and easier to dissolve when rinsed in DI water [8]. Positive photoresist is typically preferred for defining active regions in semiconductor devices because positive photoresist regions dissolve more quickly than negative photoresists. Additionally, positive photoresists have a higher resolution than negative photoresists [9].

An area where possible problems may occur is the overexposure or underexposure of the wafer. The effect of an underexposure would be that the areas of the photoresist do not receive enough of the UV light to become fully soluble and dissolve during its development. This will cause the active to contain regions that are smaller or blocked. On the other hand, an overexposure would allow the UV light to diffract beyond the desired mask boundaries, resulting more areas soluble and removed than necessary, leading to openings that are wider than intended and a larger active region [10].

After creating the desired pattern on the wafer using photolithography, etching is used to remove the silicon dioxide from the exposed areas of the wafer. This process, like thermal oxidation has a wet and dry method. Wet etching uses a liquid in order to dissolve the silicon dioxide. This method gives fast results but can lead to the removal of too much material if left for too long. On the other hand, dry etching uses plasma to remove the silicon dioxide through chemical reactions. While dry etching produces higher pattern fidelity, the equipment used is complex and may be unavailable to users.

Applications

When designing and implementing a photodiode into an electrical system, the following constraints and parameters must be taken into consideration. First, dark current is the small amount of current that flows in a photodiode circuit, even in the absence of light. This is due the base thermal energy present that contributes to the generation of electron-hole pairs. The dark current contributes to the overall noise of the system and sets the minimum threshold for detection. Responsivity is the ratio of photocurrent to incident optical power. Higher responsivity leads to increased current per unit of optical power, which improves sensitivity, though the design will still need to account for noise and dark current. Responsivity can be characterized in Equation 8.

$$R = \frac{I_{ph}}{P_{opt}} \quad (8)$$

In Equation 8, R is the responsivity, I_{ph} is the photocurrent generated by the photodiode, and P_{opt} is the incident optical power [1]. Quantum efficiency refers to the number of photons that generate collected electron-hole pairs. This measures how effectively photons are converted into charge carriers and is highly correlated with responsivity. Response time refers to how fast a current can rise and fall. Fast response time is needed for systems such as optical data systems, but less needed for devices such as street lamps. Improving one of these factors, like responsivity, can also increase dark current or noise. Thus, photodiode design must balance the different trade offs depending on its use.

Photodiodes are used in many common appliances and devices including street lamps, bank note counters and smoke detectors. Photodiodes can be used in smart street lamps to vary light intensity depending on the time of day [11]. In this use case, low dark current and moderate responsivity are needed to ensure that the lamp will be able to adjust to the changing illumination. Bank note counters contain a small laser and photodiode that counts every time a bank note breaks the laser light. In this case, fast response time is needed when designing the device in order to accurately count rapid moving bank notes. Responsivity is also needed to ensure that each bank note produces a clear electrical signal.

Photodiodes are also widely used in optical communication systems, where they detect light signals transmitted through fiber optic cables. In this application, the most important factors are response time and low junction capacitance. This is because these two parameters determine the bandwidth of the system and how quickly the device can respond to changes in light. High responsivity is also needed to ensure that even weak optical signals can be detected, especially for long transmission distances.

Finally, smoke detectors contain a similar system, when smoke fills the air and dilutes the air, the variation of light is detected, making the alarm go off. Because the smoke detector utilizes the effects of diminished light, high sensitivity and low noise are needed to detect the small changes in the light intensity.

These applications show that each semiconductor design changes depending on how the device behaves, especially how charges are created and recombined in the depletion region.

There are four main types of photodiodes that are commonly used. The p-n photodiode is the simplest device and uses a basic junction. This p-n diodes offer moderate speed and sensitivity. The PIN photodiode adds an intrinsic layer (I) between the p and n regions, which widens the depletion region and improves the quantum efficiency and response speed by allowing more photons to enter the depletion region, creating more charge carrier movement. Avalanche photodiodes operate in high reverse bias, making them highly sensitive for detecting weak optical signals. These photodiodes have the capability of even detecting a single photon. However, these devices are sensitive to noise and require precise bias control. Finally, Schottky photodiodes use a metal semiconductor junction instead of a p-n junction, allowing for extremely fast response times due to the reduced carrier transit time, but have lower sensitivity. These different device structures show the trade-offs between speed, sensitivity, and noise when designing photodiodes.

Emerging Technology

Some of the current research and emerging technology regarding photodiodes include, wide gap photodiodes, integrating photodiodes onto CMOS chips, and single photon detection.

Firstly, wide-gap or ultra wide-gap photodiodes are devices used to measure and detect high energy UV light [12]. With many applications in flame detection and space instruments, wide gap photodiodes can be implemented in extreme environment where conventional silicon diodes may fail. Creating wide gap photodiodes is done by replacing the silicone band gap with another material. An interesting property of the band gap is that using differing materials other than silicone increases performance nonlinearly, meaning that another material would not just double performance, but could increase performance by 30 times or more. However, wide-gap and ultra wide-gap materials are often difficult to work with due to tight electron bonds. This makes good crystal material harder to fabricate and electrical contacts difficult to attach.

Second, integrating CMOS chips and photodiodes together allows for high speed photodetection [13]. By integrating the CMOS and photodiode onto one chip, this allows for the conversion and amplification of optical signals on one chip, enabling faster data communication. However, high power consumption and noise prove to be challenges for researchers.

Finally, single photon detection using avalanche photodiodes is an emerging area of research [9]. Avalanche Photodiodes detect very weak light and amplify the signal internally. By biasing the photodiode just above the breakdown voltage, the photodiode can be designed to detect a single photon reliably. This can be used in long distance fiber optic communication, quantum technologies, or LiDAR and imaging. However, challenges include high noise, afterpulsing, and temperature sensitivity.

Plasma-Based Processing Concepts

Plasma-based and chemical etching are used in semiconductor fabrication to remove different layers of the material. In this lab process, oxide layers like silicon dioxide are removed during the contact etch step. Other methods such as a buffered oxide etch (BOE) or plasma etch are also used.

Objectives and Paper Organization

This paper is organized as follows. Section II describes the cleanroom environment, fabrication facilities, and tool limitations. Section III presents the full fabrication process flow, including oxidation, photolithography, etching, doping, and annealing steps. Section IV details Synopsys Sentaurus simulations, including parameter variations and doping profile analysis. Section V outlines the electrical and optical characterization methods. Section VI discusses results and comparison with expected device behavior. Finally, Section VII concludes the work and proposes future improvements.

Fabrication Resources and Process Flow

Cleanroom Environment and Safety Constraints

Due to the sensitive nature of micro and nano electronics, fabrication requires the utilization of cleanrooms. A cleanroom is a controlled environment with regulated temperature and humidity, designed to minimize the amount of airborne particles and contaminants that may affect the fabrication process.

Cleanrooms are classified by the maximum amount of allowed particles in a cubic foot of air. In this lab, Class 1000 and Class 100 clean rooms were used, meaning the Class 1000 rooms allowed for up to 1000 particles per cubic foot of air and the Class 100 rooms allowed for up to 100 particles per cubic foot of air. The Class 100 rooms are needed for specialized tasks such as photolithography, while Class 1000 rooms are used for less critical tasks such as wafer cleaning.

Cleanrooms are also kept at a slightly higher pressure than other areas, creating positive pressure. This means that when doors open, air flows out instead of in. This keeps unfiltered air and particles from entering.

Cleanrooms keep particle levels extremely low using HEPA filters, which remove at least 99.97 percent of particles that are 0.3 micrometers or larger [14]. Air is constantly circulated through these filters and then moved into the cleanroom. This flow of air prevents particles from floating around. Unwanted particles and other possible contaminants are moved away from the wafer surface and then towards vents near the floor. This helps minimize possible defects during fabrication.

Cleanrooms also use other systems to support fabrication. Deionized (DI) water is used to rinse and clean wafers because it removes ions that could interfere with device performance. DI water used in cleanrooms typically has a resistivity of 18.2 M Ω ·cm. This ensures that no unwanted ions remain on the wafer surface that could impact device performance. Gas needed for fabrication is delivered using dedicated gas cabinets that supply precise amounts of gases like oxygen and nitrogen.

In photolithography areas, yellow lighting is used to protect the photoresist. Photoresist materials are sensitive to ultraviolet and short wavelength light, and exposure to these wavelengths can unintentionally alter the pattern. Yellow lighting filters out these wavelengths, preventing unwanted exposure.

Safety Protocols

Strict safety protocols must be enforced to reduce chemical contaminants in the cleanroom. Before entering the clean rooms, gloves, hairnet, and shoe covers are donned before entering the gowning room. All electronic devices must be sprayed and wiped down with isopropanol alcohol. In the gowning room, latex gloves are worn before putting on a coverall suit. Goggles are then worn on top of the head in case of working with chemicals. All chemical handling must be done in a fume hood to prevent exposure to harmful vapors. If a chemical spill occurs, the use of eyewash stations and safety showers must be used quickly. Additionally, proper labeling, storage, and disposal of chemicals are important to ensure safety for the lab technicians and environment.

Tools and Equipment

The Lindberg Blue M furnace is available in single zone and multi zone configurations. For our lab, we used a three zone tube furnace, which allows for uniform temperature within the furnace. This is important for achieving a uniform oxide thickness [15].

The oxidation temperature range used in the Lindberg Blue M furnace ranges from 1100 degrees to 1500 degrees celsius. The ramp procedure is the slow heating and cooling of the oxidation furnace. The purpose of this is to ensure that the wafer is safely brought up to temperature. This prevents possible stress and defects from appearing on the wafer appearing due to drastic changes in temperature.

In thermal oxidation, thin oxide layers grow quickly, while thicker oxide layers grow more slowly. Thin oxides follow the linear growth regime because the surface reaction controls the rate, whereas thick oxides follow the parabolic regime because oxygen has to diffuse through the existing oxide to continue the growth [16].

Ellipsometry

Ellipsometry is a method of measuring the thickness of a film layer in a material by measuring the polarization changes in the refraction of light. This polarization change is measured with two parameters: ψ , which is the amplitude ratio, and Δ , which is the phase difference. These parameters can be used to determine film properties in the following equation.

$$\rho \equiv \frac{r_p}{r_s} = \tan^{-1}(\psi)e^{i\Delta} \quad (9)$$

In Equation 9, ρ is the ellipsometric ratio, r_p is the p-polarized complex Fresnel reflection coefficient, r_s is the s-polarized complex Fresnel reflection coefficient, ψ is the ellipsometric amplitude parameter, and Δ is the ellipsometric phase parameter. Using this equation, a light is shined onto the sample at a known angle. It measures the reflected polarization and determines the ϕ and Δ terms.

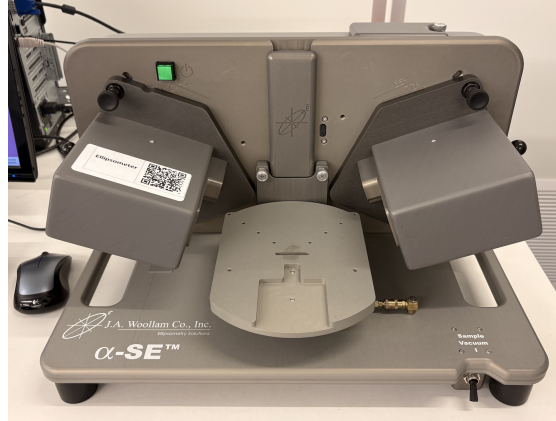


Figure 4: Woollam Alpha-SE ellipsometer used in cleanroom for photodiode fabrication.

A Woollam Alpha-SE ellipsometer, see Figure 4, was used to measure the thickness of the thermally grown SiO_2 layer, measuring at 285.3 nm (2853 Å). This thickness falls within the expected range of 300 nm (3000 Å) for thermal oxidation.

Scanning Transmission Electron Microscope

A Scanning Transmission Electron Microscope (STEM) is an advanced imaging device that uses electron beams to visualize the structure of materials at a nanometer scale. The STEM combines the features of a Transmission Electron Microscope (TEM) and a Scanning Electron Microscope (SEM). A TEM device utilizes a beam of electrons through a very thin sample in order to examine its internal atomic structure. The sample, typically smaller than 100 nm, is placed in a large vacuum chamber, and then exposed to a high energy electron beam. Because electrons have a smaller wavelength than visible light, it is able to have a higher microscope resolution. On the other hand, a SEM device uses a focused electron beam to scan the surface of a sample to produce detailed surface images. These samples can be of any thickness due to the fact that the electrons only hit the surface of the sample. Generated images can have a resolution from less than a nanometer to a few nanometers in size. STEM is a combination of TEM and SEM capabilities and technology. A STEM integrates the SEM's focused electron beam features and the TEM's internal imaging capabilities, allowing the STEM to reach a higher imaging resolution of a material's internal structure, down to the sub-angstrom level. See Figure 5.

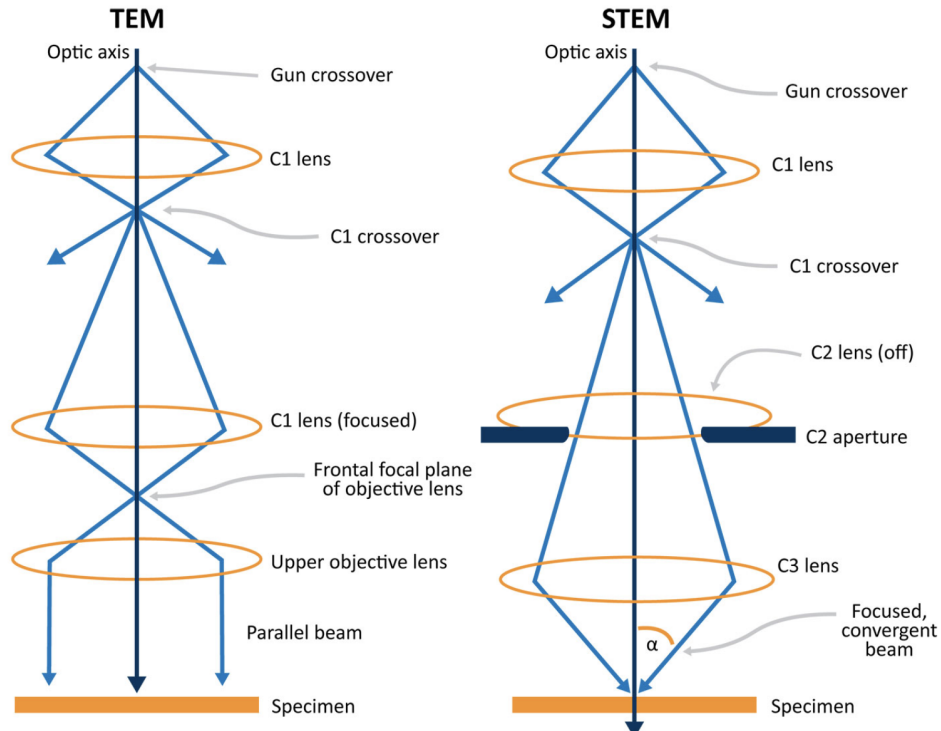


Figure 5: Schematic diagram of beam curves of the TEM and STEM. [17]

The STEM works by generating a high energy electron beam, typically from 60-300 keV, onto an extremely small, atomic-scale, probe. The beam then scans the sample, point by point, across its length. From the scattering of electrons, an image can be reconstructed off the detected signals. A variety of samples can be examined using this device, such as semiconductor cross sections (Si, GaAs, etc.), thin films, and oxide layers, perfect for semiconductor fabrication and nanomaterials research. This is especially an advantage over an Atomic Force Microscope (AFM) due to the fact that it can examine the structure of a sample without physical tapping or continuous contact with the sample with a needle.

Some of the strengths of using a STEM instead of other imaging devices include its extremely high resolution without physical interaction with the sample, and its ideal fit for evaluating semiconductor materials with its ability to examine dopant distribution, oxide thickness, and p-n junction sharpness. However, due to the electron beams' high energy, the focused beam can potentially damage the sample by moving atoms or altering softer materials. Additionally, samples must be carefully selected, needing to be 100nm or less in thickness in order to achieve electron transparency. This tool would be especially useful in our lab course, as we fabricate a photodiode semiconductor device. Its uses could include investigating the dopant distribution, analyzing the oxide thickness, or examining defects in the wafer. At WashU, the STEM, also known as the Electron Beam Lithography device, is located in the Institute of Materials Science and Engineering (IMSE) department and used by various labs and departments in physics and engineering. Uses of this device including imaging small creatures, like bugs, or even etching drawings onto samples at a microscopic scale.

Spin Coating

Spin coating is one of the most common ways of applying photoresist onto silicon wafers in the photolithography process. Spin coating is often chosen due to its uniform, controlled, and repeatable process. Despite this, spin coating has a few glaring limitations.

First, spin coating only works on flat plane surfaces. When wafers have grooves or a 3d surface, it becomes difficult for the spin coat to cover all surfaces. This may lead to variations in coating thickness and other defects such as striations or lack of coating in certain areas. Therefore, uniform photoresist layers are reserved for flat wafers [18].

Second, spin coating depends on the RPM of the spin. Small variations of the spin speed and acceleration can affect the final thickness of the photoresist or other deposited film. Spin coating also results in much of the photoresist being wasted. During the process, most of the photoresist is thrown off the wafer as it is spun, leaving only a small portion left on the wafer.

Finally, many defects can occur in spin coating. Even small particles can lead to marks in the spin coat process, reducing uniformity and may affect the quality of the pattern in photolithography. [19].

To address some of these limitations. Many alternative methods of photoresist deposition exist. First, spray coating

is the process of using an ultrasonic spray nozzle to deposit photoresist as small droplets. Electrodisposition involves the process of driving photoresist particles into the wafer through an electric field. Drip coating is when a substrate is immersed in a photoresist solution and then slowly withdrawn, leaving behind a thin film [20]. A meniscus coating is when photoresist is deposited as a liquid meniscus between the coating and the substrate [21]. This forms a uniform film as the substrate moves underneath. Finally, a curtain coating is when a continuous curtain of liquid photoresist flows onto a moving wafer, creating a uniform coating across large surface areas [22].

See Table 1 for more comparison.

Deposition Method	Principle of Operation	Key Characteristics	Advantages	Disadvantages
Spray Coating	Uses an ultrasonic spray nozzle to deposit photoresist as small droplets	Works well on 3D surfaces and substrates with cavities; thickness is controlled by spray conditions and liquid viscosity	Provides good coverage of 3D structures, reduces photoresist waste, and uniformly coats irregular surfaces	Requires specialized spray equipment; variations in cavity geometry can cause thickness nonuniformity
Electrodisposition	Drives photoresist particles onto the wafer through an electric field	Produces uniform coatings in deep structures; thickness is controlled by voltage and deposition time	Coats sidewalls effectively and enables high-aspect-ratio features	Requires a conductive surface layer, specialized materials, and additional processing steps
Dip Coating	Substrate is immersed in a photoresist solution and then slowly withdrawn, leaving behind a thin film	Film thickness is determined by withdrawal speed, photoresist viscosity, and solvent evaporation rate	Simple process that can produce uniform coatings and is suitable for large substrates	Limited control on complex surfaces; slower process that may produce edge effects and thickness variation
Meniscus Coating	Photoresist is deposited as a liquid meniscus between the coating head and the substrate as the substrate moves underneath	Continuous coating method with good thickness control; best suited for large-area substrates	Efficient material usage, uniform large-area coatings, and scalability for industrial production	Requires precise control of fluid flow and substrate motion; less suitable for complex 3D topography
Curtain Coating	A continuous curtain of liquid photoresist flows onto a moving wafer, creating a uniform coating across large surface areas	Can process many substrates at once; thickness is controlled by flow rate and substrate speed	Enables fast coating over large surfaces with efficient material usage and good thickness uniformity	Requires stable liquid flow, complex equipment, and is not well suited for small substrates or 3D objects

Table 1: Comparison of alternative photoresist deposition methods.

Of the five alternative methods for photoresist deposition, spray coating would be the best method for coating the patterned wafers used in this lab. Because the droplets land where they are deposited, the resist can cover more uneven surfaces and cover features such as grooves or cavities better. Additionally, spray coating wastes less material because the resist is deposited only where it is needed instead of being spun off the wafer edges. Thus, because the wafers used contain patterned features, using a spray coating would not only be more resource efficient compared to spin coating, but also create a more uniform covering of the surface compared to other deposition methods.

Materials and Substrates

The photodiodes were fabricated on 4-inch silicon wafers with $\langle 100 \rangle$ crystal orientation, approximately 500 μm thickness, and a background doping concentration of about 10^{15} cm^{-3} . Boron implantation was used to form the p-type region of the device. Silicon is used widely in the industry due to its good optical properties, low cost, and easy maneuverability [23]. Additionally, because silicon has a bandgap energy of about 1.12 eV at 300K [4], this means that its cutoff wavelength is about 1100nm, meaning that photodiodes will be responsive to visible and infrared light. A SiO_2 layer was thermally grown on the wafer, utilizing the thermal oxidation process. The oxide layer is used as a mask when dopants are introduced, provides insulation between different device regions, passivates the silicon surface, and serves as a dielectric [24]. Additionally, the silicon wafers were doped with boron, creating a p-type substrate. The p-type substrate was chosen in order to create the p-n junction within the material, creating a depletion region suitable for charge carriers.

Wafer Preparation and Initial Oxidation

Before the wafer was processed, it was rinsed thoroughly using acetone, isopropyl alcohol, and DI water. Then, the wafer was dried with nitrogen. Even small surface imperfections can lead to an increase in dark current and negatively affect the depletion region. Cleaning the wafer allows for a uniform photoresist coating and thus better lithography results. Thermal Oxidation was then conducted on the wafer following the process outlined in the Fabrication Process using a Lindberg Blue M furnace. This process was done at 1100°C for approximately 10 hours to achieve the target oxide thickness.

Photolithography and Pattern Transfer

Photolithography is used to transfer patterns from a photomask onto the silicon wafer. This process defines where a material will be added or removed. This process starts by spin-coating a thin, uniform layer of photoresist onto the wafer, then soft baking on a hot plate to remove solvents and improve the bond between the photoresist and the wafer. After this, a photomask with the desired pattern is then aligned over the wafer and then exposed to ultraviolet light.

Since a positive photoresist was used, the exposed regions become more soluble and then are removed using a developer, leaving behind the desired pattern in the remaining resist. This pattern created by the photoresist then acts as a mask for later steps, such as etching or ion implantation.

In addition to UV exposure and development, a descum step was performed using a brief oxygen plasma clean to remove any remaining photoresist in the patterned regions. This helps ensure that the openings are fully cleared before etching and improves the accuracy of the transferred pattern.

After development, the wafer was also subjected to a hard bake to further strengthen the photoresist and improve its adhesion to the surface. This makes the resist more resistant to plasma etching and helps prevent it from lifting during later steps.

Photolithography alignment and UV exposure were carried out using a UV-KUB 3 mask aligner. This tool provides a UV exposure at 365 nm and allows alignment accuracy to a 1 μm precision, which is very useful in small device features [25].

Correct alignment and accuracy in the photolithography process is important because it determines the size and placement of device features, such as contact openings and active regions. Any slight misalignment can cause defects that negatively impact the device.

Etching Processes

Etching is used to remove material from the wafer surface to form features on the wafer such as contact openings. In this process, reactive ion etching (RIE) was used to remove silicon dioxide while still keeping silicon underneath. RIE uses both chemical and physical methods. A reactive plasma chemically reacts with the oxide, while ion bombardment physically removes material from the surface.

RIE utilizes a negative self-bias voltage on the wafer, which moves positive ions toward the surface. This results in anisotropic etching, where the material is removed in the vertical direction. As a result, the process produces well defined features with straight sidewalls, which is important in forming contact regions in the photodiode.

The RIE process utilized CHF_3/Ar gas chemistry, with chamber pressure on the order of tens of mTorr and RF power in the range of 100–300 W. These parameters control etch rate, anisotropy, and selectivity.

Etch selectivity is important because the SiO_2 must be completely removed without damaging the underlying silicon. Over-etching can cause surface damage and decrease performance. The SiO_2 etch rate for our process was approximately 2000 $\text{\AA}/\text{min}$, and the required etch time was calculated using the oxide thickness in both the active and field regions.

Following the etching step, a buffered oxide etch (BOE) clean is performed to remove any thin native oxide that forms when silicon is exposed to air. A BOE is a chemical solution that is used to remove silicon dioxide. It is usually

a combination of hydrofluoric acid and a buffer like ammonium fluoride. The buffer helps control the etch so it's not too aggressive. The solution works by reacting with the oxide and then dissolving it evenly. A short BOE is performed before aluminum deposition in order to remove that thin native oxide layer when exposed to air. If it is not removed, it may lead to poor electrical contacts and higher resistance. This is because the oxide layer acts like an insulating layer between the aluminum and the silicon.

Doping and Junction Formation

Wafers were sent off site to CuttingEdge Ions, LLC and implanted with $^{11}\text{B}^+$ ions (boron), with a dose of 10^{13} cm^{-2} , at an energy of 100 keV, a tilt angle of 7° , and wafer rotation during implantation. The implantation energy will determine the projected range, or R_p , of dopant ions in the silicon lattice. At an implantation energy of 100 keV, the boron atoms will be able to penetrate the surface, following a gaussian distribution. See Equation 10.

$$N(x) = N_{max} \exp\left(-\frac{(x - R_p)^2}{2\sigma^2}\right) \quad (10)$$

In Equation 10, $N(x)$ is the dopant concentration at a depth x , N_{max} is the peak dopant concentration, x is the depth below the surface, R_p is the projected range, or the average penetration depth of the ions, and σ is the standard deviation of ions around R_p .

The peak dopant concentration N_{max} is related to the implantation dose Q and can be characterized in the following equation.

$$N_{max} = \frac{Q}{\sqrt{2\pi}\sigma} \quad (11)$$

In Equation 11, N_{max} is the peak dopant concentration, Q is the implantation dose, and σ is the standard deviation of the implanted ion distribution. This equation shows that higher implantation dose leads to higher peak concentration. This then affects the electric field strength and junction formation within the device.

The implantation dose of 10^{13} cm^{-2} gives the peak dopant concentration and influences the junction depth and electric field strength. This dosage was chosen to ensure that a junction capable of creating charge carriers could be formed. Increasing the dose makes the electric field stronger in the depletion region. However, if the dose is too high, it can also lead to more recombination and increased leakage current.

A tilt angle of 7° was also implemented to ensure the more boron atoms would be implanted into the silicon lattice structure. Due to the uniformity of the lattice structure, it is possible for boron to pass straight through an area without being embedded into the material, called channeling. This can lead to a non-uniform doping profile and junctions that are deeper than desired. By tilting the wafer slightly, this disrupts these paths, increasing collisions, and helps the dopants become embedded more consistently. As a result, the tilt improves uniformity and control of the doping profile.

Annealing and Thermal Processing

After ion implantation, the wafers go through a thermal annealing process in order to activate the dopants and also repair damage done to the crystal. Annealing is a necessary step because many dopant atoms end up in the wrong spots during implantation and are electronically inactive. Annealing at high temperatures allows these atoms to move into the right lattice positions, where they can then generate free carriers, becoming electronically active. It also helps fix defects caused by ion impacts.

Annealing also causes some dopant diffusion. Because of this, the junction depth depends on the initial implantation and the annealing step. Controlling the annealing temperature and time is important in order to ensure device performance. Too much diffusion can decrease the electric field and increase leakage current.

BEOL Processes

After the junction is formed through ion implantation and annealing, back end of line processing is needed to make the photodiode usable as an electrical device. Even if the pn junction is formed correctly, the device still needs reliable insulation, contact openings, and metal wiring so that it can be connected and tested.

A typical BEOL process starts by adding an insulating dielectric layer, which helps separate conductive regions and prevents short circuits. Contact openings or vias are then etched through the dielectric so that metal can reach the semiconductor underneath. After this, metal is deposited and patterned to form the wiring and contact pads. In more advanced processes, chemical mechanical planarization may also be used to polish and flatten the wafer surface before additional layers are added.

For this photodiode, the intended BEOL process included opening the contact regions, removing native oxide, depositing aluminum, and patterning the metal. The native oxide removal step is important because exposed silicon naturally

forms a thin oxide layer when it comes into contact with air. Since silicon dioxide is insulating, it can increase contact resistance and make electrical measurements less accurate. A short buffered oxide etch before aluminum deposition helps remove this oxide and allows the aluminum to form a better contact with the silicon.

After the metal contacts are formed, the wafer must be prepared for dicing and packaging. Before dicing, a protective photoresist layer can be applied to the wafer surface to reduce scratches and contamination from cutting debris. The wafer is also mounted on dicing tape so that the individual chips stay in place after separation. During dicing, DI water is used for cooling and cleaning because tap water can leave minerals or ionic residue on the wafer. These contaminants could cause corrosion, leakage paths, or reduced device performance.

Packaging is especially important for photodiodes because the device must be protected while still allowing light to reach the active area. Unlike a purely electrical device, a photodiode package needs an optical path for incoming light. Commercial photodiode packages often use a clear glass window to protect the device from dust and moisture while allowing light to enter. Some packages also use microlenses or careful optical alignment to focus light onto the active region. If the window or incoming beam is misaligned, the measured photocurrent can decrease even if the pn junction itself is working properly.

Deviations from Intended Process Flow

A major issue occurred during the oxide etching step, where too much material was removed and the silicon dioxide layer was completely etched away, leaving bare silicon exposed. This suggests that the photoresist mask did not properly protect the oxide. One likely reason is poor photoresist processing, such as uneven or insufficient hard baking, which can weaken its ability to stick and resist etching. As a result, the photoresist may have lifted or broken down during the plasma etch, allowing oxide to be removed where it should not have been. Changes in etch time or plasma conditions could have made the problem worse.

This is a serious problem because removing the oxide layer gets rid of electrical isolation and makes the wafer unusable. Without the oxide, the device can not be properly formed. This shows how important it is to carefully control both the photoresist and etching steps, since mistakes in either one can permanently damage the wafer.

Facility and Tool Limitations

The fabrication process was limited by equipment availability. In particular, the PECVD system was down due to a power outage, which prevented the deposition of thin films needed for later steps.

Overall Process Flow Diagram

Figure 6 uses a simplified labeled process flow.

Synopsys Sentaurus Simulations

Synopsys Sentaurus is a TCAD (Technology Computer-Aided Design) software used to simulate semiconductor fabrication processes. Sentaurus Workbench was used to organize and manage the simulation. SProcess was used to model the fabrication steps to generate the device structure. SDevice is used to simulate the electrical behavior of the photodiode, such as I-V characteristics. SVisual is used to analyze and visualize the simulation results, including doping profiles and electric field distributions. Using the Linux cluster provided by WashU, every step of the fabrication process outlined in Figure 6 was simulated. The simulation was updated to focus on the actual photodiode fabrication process rather than only the earlier implantation-energy assignment. The main simulated process used dry thermal oxidation at 1100°C for 10 hours, boron implantation at 100 keV with a dose of $1 \times 10^{13} \text{ cm}^{-2}$, a 7° implant tilt, and annealing at 1100°C for 480 minutes.

Using provided code samples, the simulation grid was defined and silicon substrate area was created. Then the substrate was initialized to orient the crystals and doping. Thermal oxidation was used to create an SiO_2 layer. After, a layer of photoresist was added before oxide etching. See Figure 7.

The final simulated photodiode structure is shown in Figure 8. This structure represents the completed SProcess flow, including oxidation, lithography, oxide etching, boron implantation, nitride deposition, annealing, contact opening, aluminum deposition, and metal patterning. The color scale shows the net active doping concentration in the device.

This final SProcess structure was then used as the input for the SDevice electrical simulation.

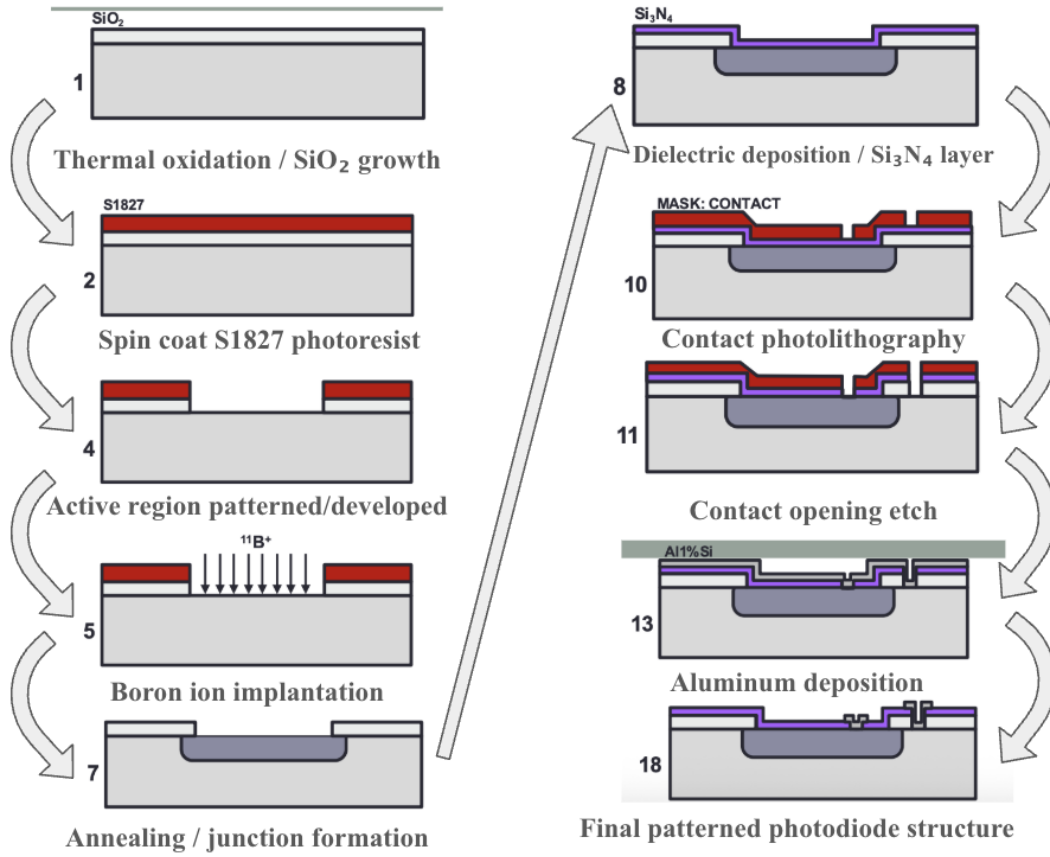


Figure 6: Simplified process flow for silicon photodiode fabrication.

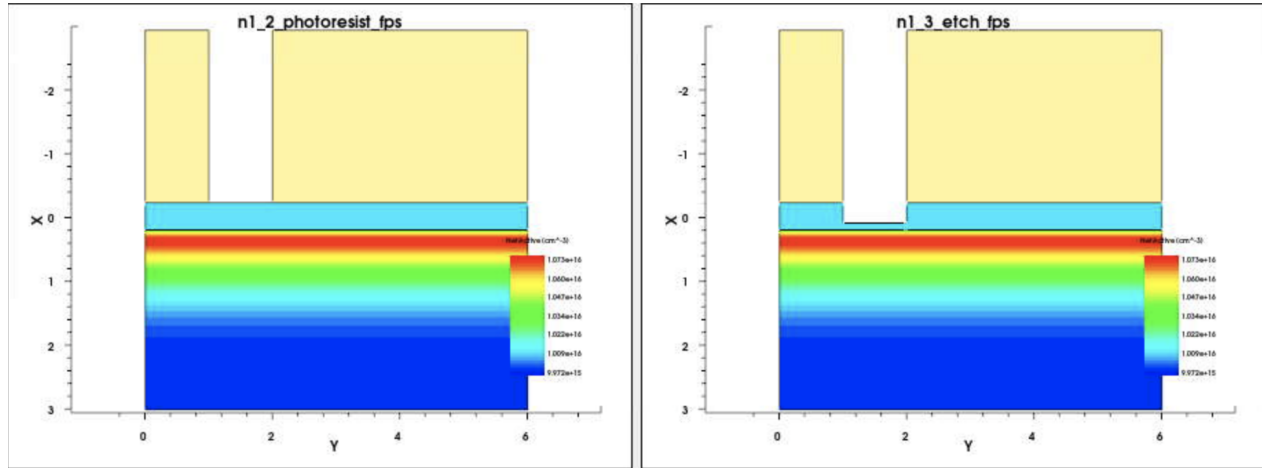


Figure 7: Synopsys Sentaurus simulation of current fabrication process.

Simulation Parameter Variations

The earlier simulation work compared different implantation energies to understand the effect of energy on dopant depth. For the final paper, the focus was shifted to the actual process parameters used for the photodiode. The main device simulation used the 100 keV boron implantation case because this matched the lab process.

To investigate the impact of fabrication parameters on device behavior, multiple simulation cases were performed by varying the ion implantation energy while keeping other parameters constant. In addition to the baseline case (100 keV), simulations were conducted at 10 keV and 1000 keV to see how implantation energy affects dopant distribution and junction formation.

All simulations used a boron implantation dose of $1 \times 10^{13} \text{ cm}^{-2}$ and were followed by annealing at 1100°C for 480

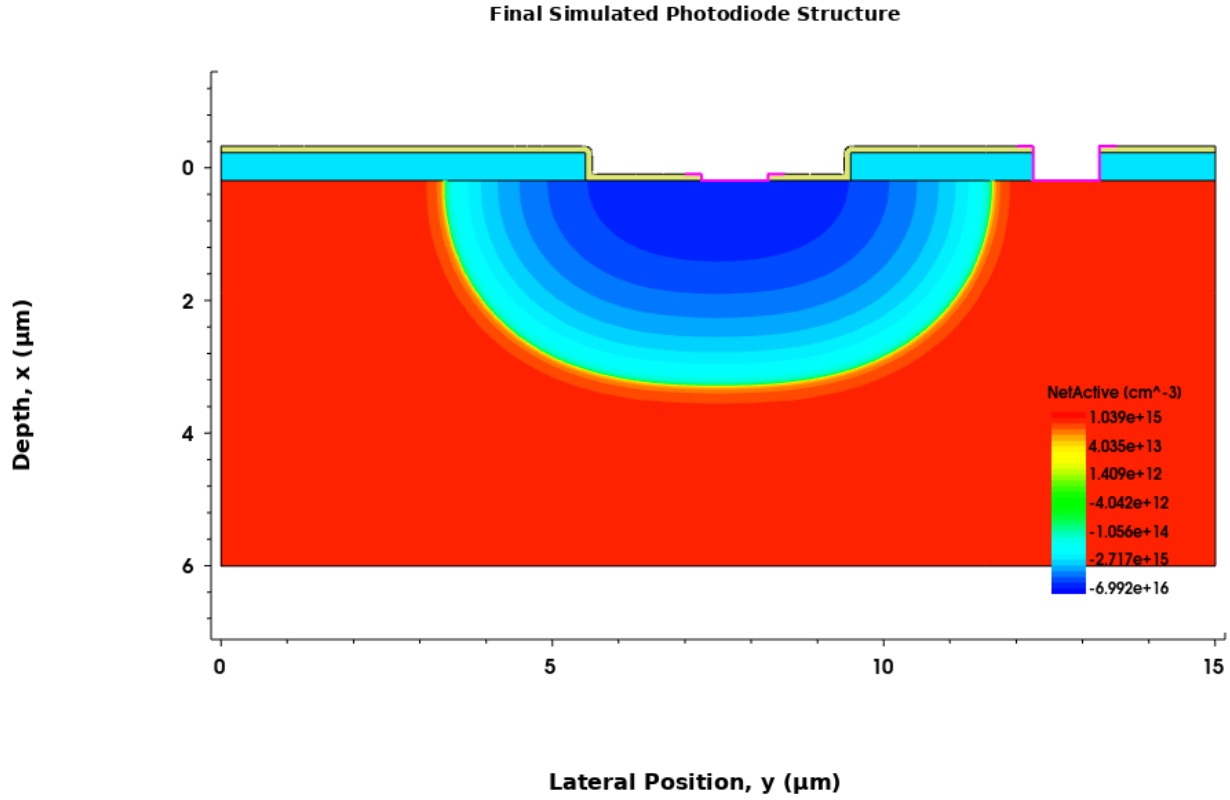


Figure 8: Final simulated photodiode structure from Sentaurus SProcess.

minutes.

Doping Profile Analysis

Doping concentration profiles were extracted from the simulations to analyze how dopant distribution varies with depth. The as-implanted profiles exhibit a Gaussian-like distribution centered around the projected range of the implanted ions.

After annealing, the dopant profiles broaden due to diffusion, resulting in a reduced peak concentration and increased junction depth. Lower implantation energies (10 keV) produce shallow, high-concentration profiles near the surface, while higher energies (1000 keV) result in deeper dopant penetration and wider distributions.

For the actual photodiode process, the 100 keV boron implant was analyzed before and after annealing. Figure 9 shows the as-implanted boron profile, the active boron profile after annealing, and the background phosphorus concentration.

As shown in Figure 9, the as-implanted boron profile has a higher and narrower peak. After annealing, the active boron profile spreads deeper into the silicon and has a lower peak concentration. This is expected because high-temperature annealing activates the dopants and causes diffusion. The resulting dopant profile affects the junction depth, depletion region, junction capacitance, leakage current, and overall photodiode behavior.

Quantitative Comparison

Table 2 summarizes the effect of implantation energy on dopant distribution.

Table 2: Effect of implantation energy on dopant profile and junction depth

Energy (keV)	Peak Concentration (cm ⁻³)	Projected Range (μm)	Junction Depth (μm)
10	2.07×10^{18}	0.199	2.14
100	4.34×10^{17}	0.497	2.25
1000	3.46×10^{17}	2.00	3.61

The results show that implantation energy strongly influences dopant distribution and junction depth. Lower energies produce higher peak concentrations near the surface but shallower junctions, while higher energies result in deeper junctions with reduced peak concentration.

Dopant Profile Before and After Annealing

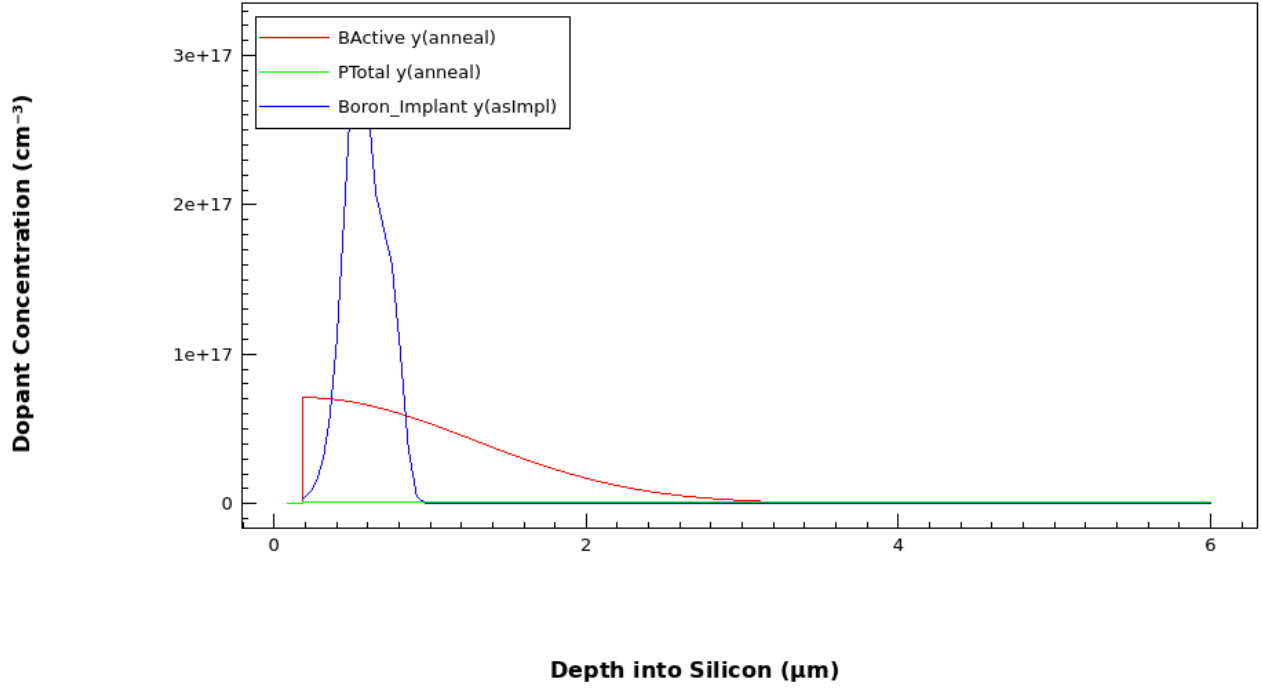


Figure 9: Simulated dopant concentration profile before and after annealing. Blue curve shows the as-implanted boron profile. Red curve shows the active boron profile after annealing. Green curve shows the background phosphorus concentration.

Annealing further increases junction depth by allowing dopants to diffuse and become electrically active. This demonstrates a key trade-off in fabrication: achieving high surface concentration versus controlling junction depth.

These variations directly impact photodiode performance, as junction depth and doping concentration influence depletion width, junction capacitance, and device response speed.

The final simulated structure was also imported into SDevice for electrical characterization. A dark I-V simulation was performed by holding the cathode at 0 V and sweeping the anode from 0 V to -5 V. This represents the reverse-bias condition of the photodiode without illumination.

As shown in Figure 10, the simulated dark current remains very small as reverse bias increases. At an applied reverse bias of -5 V, the current is approximately 4.0×10^{-15} A. This low current is expected because the simulation represents an idealized device. In the real fabricated device, defects, surface damage, contamination, poor contacts, or oxide over-etching could increase the leakage current.

A simplified illuminated SDevice simulation was also attempted by adding an optical generation term, but the illuminated simulation did not converge reliably enough to use as a final quantitative result. Because of this, the completed SDevice result included in this paper is the dark I-V simulation. This still connects to optical characterization because dark current is the baseline needed to calculate photocurrent. In an illuminated measurement or simulation, the photocurrent is calculated as

$$I_{ph} = I_{light} - I_{dark}. \quad (12)$$

The photocurrent can then be used with incident optical power to calculate responsivity.

Device Characterization Methods

Device characterization is needed to determine whether the fabricated photodiode behaves like a working p-n junction and whether it responds to light. The two main types of testing are electrical characterization and optical characterization. Electrical testing evaluates the I-V behavior in the dark, while optical testing compares the current response before and after illumination.

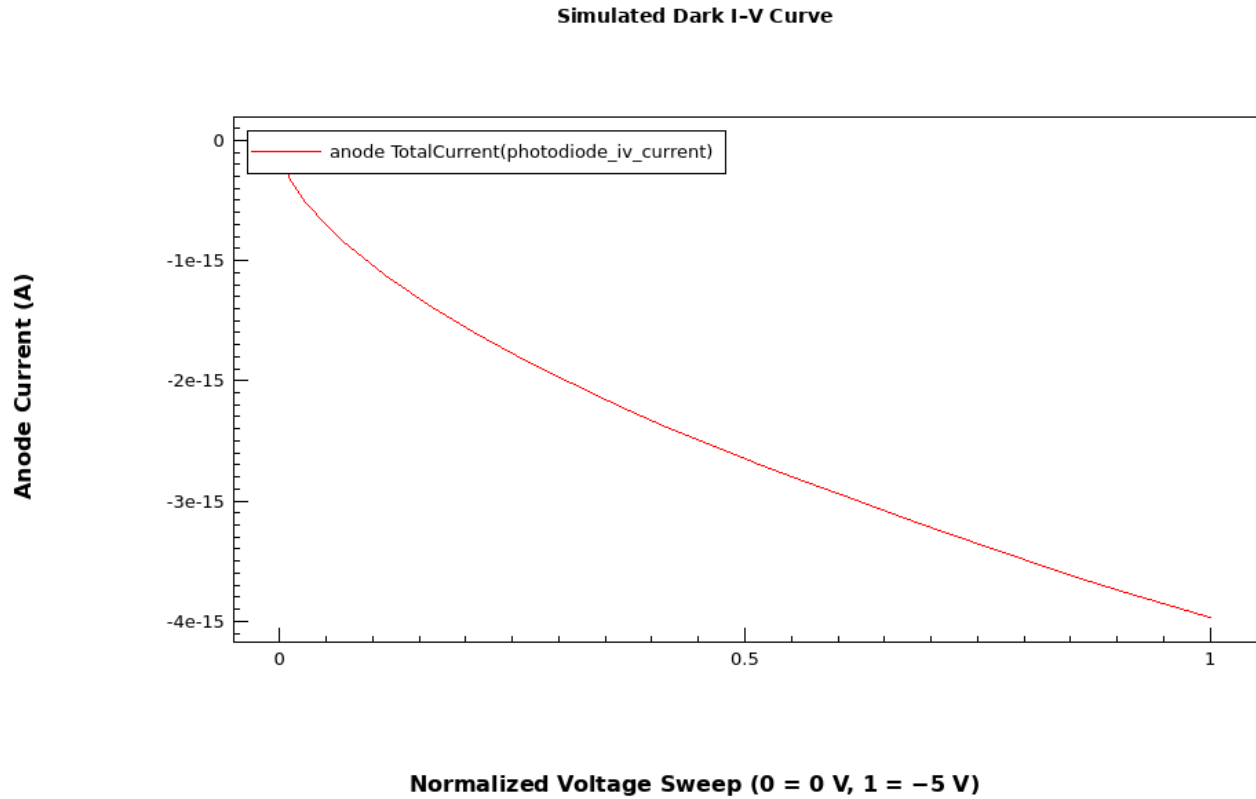


Figure 10: Simulated dark I-V curve of the photodiode using SDevice. X-axis represents the normalized voltage sweep, where 0 corresponds to 0 V and 1 corresponds to -5 V.

Measurement Setup and Instrumentation

The photodiode can be tested using a probe station connected to a semiconductor parameter analyzer or source-measure unit (SMU). The probe station allows small probe tips to contact the metal pads of the device, while the SMU applies a voltage and measures the resulting current. For dark current measurements, the device should be covered or placed in a dark enclosure so that room light does not affect the current. For optical measurements, a controlled light source, such as an LED or laser, can be directed onto the active area of the photodiode. If available, an optical power meter should be used to measure the incident optical power so that responsivity can be calculated.

The photodiode characterization setup uses a probe station, a source measure unit (SMU), a calibrated reference photodiode, an optical power meter, and laser pointers of different wavelengths. The probe station is used to contact the small metal pads of the photodiode without permanently packaging the device. It contains a stage or vacuum chuck to hold the wafer or chip, probe manipulators to position the probe tips, and a microscope to view the contact pads during alignment.

Electrical measurements are performed using a Keysight B1500A Semiconductor Device Analyzer. This tool functions as an SMU, meaning that it can apply a voltage and measure current at the same time, or apply a current and measure voltage. This is useful for I-V characterization because the voltage can be swept while the resulting current is recorded. A current compliance limit should be set before testing the photodiode for the first time. This protects the device by limiting the maximum current that can flow if the device is shorted or damaged.

For optical testing, a Thorlabs PM100D2 optical power meter and a calibrated reference photodiode can be used to determine the optical power incident on the device. The calibrated reference photodiode is first placed at the same position where the fabricated photodiode will be tested. This allows the optical power to be measured before the fabricated device is placed in the beam path. This step assumes that the light intensity and beam position stay the same when the reference photodiode is replaced with the fabricated device.

Care is also needed when landing the probe tips on the device. The probes should be lowered slowly and should not be dragged across the surface after contact is made. Dragging the tip can scratch the metal pad or damage the device. Probe tip material and radius also matter. Tungsten probe tips are commonly used because they are hard and provide stable electrical contact. If the tip radius is too large, it can short nearby contacts, while a very small tip can concentrate force onto the pad and increase the risk of damage.

Electrical characterization was performed using a probe station connected to a Keysight B1500A Semiconductor

Device Analyzer. The SMU was connected to the probe manipulators, and the probe tips were carefully placed on the device electrodes before running the measurement program. The B1500A was used to apply voltage and measure the current through the device. The exact voltage sweep range, voltage step size, and current compliance limit should be confirmed from the saved SMU program or lab notes. If these values cannot be found, they should be described as proposed measurement settings rather than confirmed experimental settings.

Optical characterization was performed using a calibrated photodiode connected to a Thorlabs PM100D2 optical power meter. This was used to measure the optical power from the light source before testing the photodiode. Measurements were taken at different wavelengths, including 405 nm, 520 nm, and 635 nm. For one comparison device, the measured responsivity values were 0.0112 A/W at 405 nm, 0.0226 A/W at 520 nm, and 0.013 A/W at 635 nm. One source of uncertainty was that the measured optical power could change depending on the position and distance of the calibrated photodiode from the light source.

Electrical Characterization Methods

Electrical characterization is done by sweeping the applied voltage and measuring the current through the photodiode. In forward bias, the current should increase quickly once the diode begins to turn on. In reverse bias, the current should stay very small because only a small leakage current flows when there is no light. This leakage current is called the dark current.

The dark I-V curve is useful because it gives information about the quality of the pn junction and the fabrication process. A low reverse-bias current usually means the junction has good isolation and low leakage. A high reverse-bias current may suggest defects, surface damage, poor junction formation, or process issues such as over-etching. Since our fabrication process had an oxide over-etch issue, the completed device may have shown higher leakage current or less ideal diode behavior.

The measured current can be compared to the ideal diode behavior using the Equation 6. In an actual fabricated device, the I-V curve may not match the ideal equation perfectly because of series resistance, recombination, leakage paths, contact resistance, and fabrication defects.

Optical Characterization Methods

Optical characterization is used to check whether the photodiode produces current when it is exposed to light. First, the current is measured in the dark to find the baseline dark current. Then, the device is exposed to a controlled light source and the current is measured again. If the photodiode is working properly, the current under illumination should be larger than the dark current because incoming photons generate electron-hole pairs in the silicon.

The photocurrent can be found by subtracting the dark current from the illuminated current,

$$I_{ph} = I_{light} - I_{dark}, \quad (13)$$

where I_{ph} is the photocurrent, I_{light} is the current measured under illumination, and I_{dark} is the current measured without illumination. This subtraction is important because it separates the current caused by light from the leakage current of the device.

If the optical power reaching the device is known, the responsivity can be calculated using Equation 8. A higher responsivity means the device produces more current from the same optical power.

The optical response also depends on wavelength because silicon only absorbs photons with enough energy to generate electron-hole pairs. The laser spot size and alignment are also important. If the laser spot is larger than the active area, only part of the optical power reaches the photodiode. This can make the measured photocurrent smaller and make the calculated responsivity appear lower than the true value. For this reason, the laser should be carefully aligned under the microscope, and the active area should be placed as close as possible to the same position used during optical power calibration.

The maximum possible responsivity can be estimated by assuming an external quantum efficiency of 1. For a wavelength in nanometers, the ideal responsivity is approximated as

$$R_{max} = \frac{\lambda}{1240}, \quad (14)$$

where R_{max} is the maximum responsivity in A/W and λ is the wavelength in nm. For a 650 nm red laser,

$$R_{max} = \frac{650}{1240} \approx 0.524 \text{ A/W}. \quad (15)$$

For a 450 nm blue laser,

$$R_{max} = \frac{450}{1240} \approx 0.363 \text{ A/W}. \quad (16)$$

This shows that the ideal responsivity is higher at longer wavelengths. Longer-wavelength photons have lower energy, so for the same optical power, more photons arrive at the device each second. If those photons are absorbed and collected efficiently, they can generate more electron-hole pairs and produce a larger photocurrent.

Although the main fabricated wafer was over-etched and did not give reliable final photodiode results, optical characterization was still performed using available comparison devices. For Swojan’s wafer, Die 10 showed a measurable response when exposed to different wavelengths of light. The highest responsivity was measured with green light at 520 nm, while the purple and red light produced lower responsivity values. This shows that the device was able to generate photocurrent under illumination, although the measured responsivity was much lower than the ideal responsivity predicted by $R_{max} = \lambda/1240$.

Table 3: Measured responsivity for Swojan’s wafer, Die 10.

Color	Wavelength (nm)	Responsivity (A/W)
Purple	405	0.0112
Green	520	0.0226
Red	635	0.0130

Another tested device, Die 14, did not show stable photodiode behavior. Photocurrent mainly appeared below approximately -6 V , and the measured current was unstable. This suggests that the device may have behaved more like a Schottky-like metal-insulator-semiconductor contact rather than an ideal p-n photodiode. The unstable current may have also been caused by defects or trapped states at the surface or interface.

Under illumination, the photodiode current can be written as

$$I = I_s \left(e^{\frac{qV}{nkT}} - 1 \right) - I_{ph}, \quad (17)$$

where I_s is the reverse saturation current, q is the elementary charge, V is the applied voltage, n is the ideality factor, k is Boltzmann’s constant, T is temperature, and I_{ph} is the light-generated photocurrent. Compared to the dark I-V curve, the illuminated curve shifts downward because absorbed photons produce additional current in the photodiode.

Results and Discussion

Table 4: Criteria for evaluating photodiode functionality.

Criterion	Good Device	Failed Device
Dark current	Low reverse-bias current with no light	High leakage current in reverse bias
I-V behavior	Clear rectification with low reverse current and higher forward current	Straight line through origin, indicating resistor-like behavior or shorting
Photocurrent	Current increases when light is applied	Little or no difference between dark and illuminated current
Responsivity	Reasonable photocurrent per optical power	Very low responsivity, showing poor light-to-current conversion

The illuminated I-V curves for Swojan’s wafer, Die 10, also showed photodiode-like behavior. As shown in Figure 11, the illuminated curves produced larger reverse-bias current than the dark curve. This indicates that the device generated photocurrent when exposed to light. However, the response was still much lower than what would be expected from a high-quality commercial silicon photodiode, suggesting that carrier collection was limited by contact quality, surface recombination, or process-related defects.

We measured the dark I-V curve to verify the device’s basic diode characteristics prior to illumination. The most successful photodiode (2025 Die 7) exhibited a low dark current of approximately 0.984 nA at -1 V . This minimal baseline leakage is critical, as it reduces background noise and improves the device’s overall sensitivity to light.

While the device demonstrated clear rectifying behavior with a ratio of 5.875, its performance deviated from an ideal p-n junction. Specifically, the high extracted ideality factor of 17.14 (measured between 2 V and 3.5 V) indicates the presence of non-ideal mechanisms. These deviations are likely driven by real-world fabrication factors such as contact resistance or surface leakage.

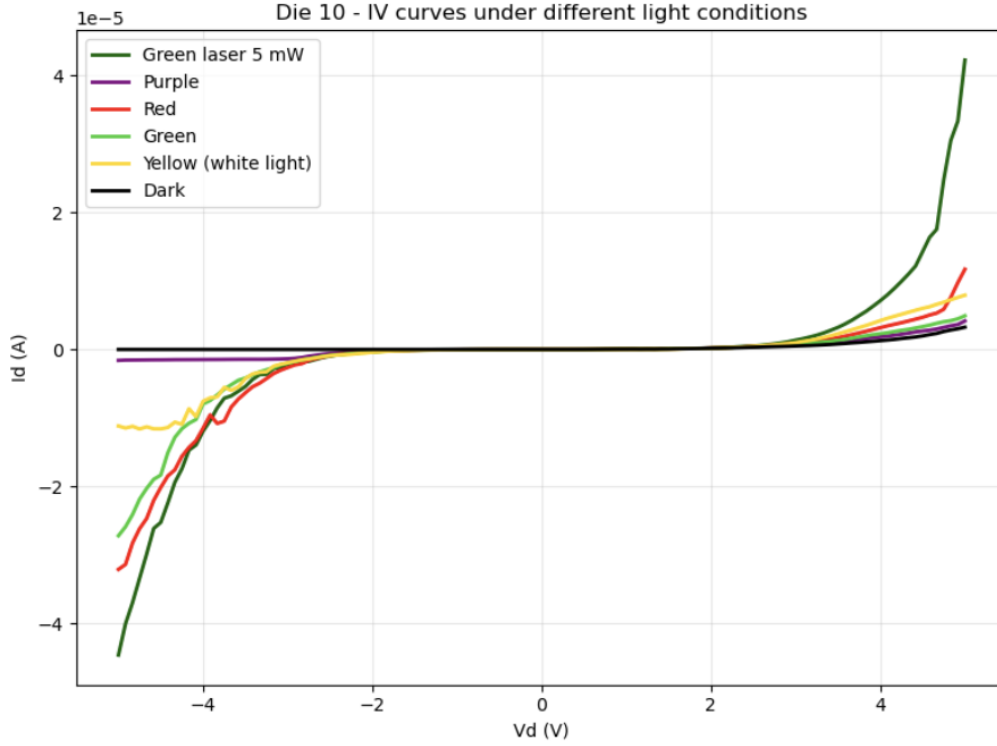


Figure 11: Measured I–V curves for Swojan’s wafer, Die 10, under different light conditions. The illuminated curves show increased reverse-bias current compared to the dark curve, confirming photocurrent generation.

Comparison with State-of-the-Art Semiconductor Fabs

The GlobalWafers MEMC visit also showed how different industrial wafer manufacturing is compared to a teaching cleanroom [26]. The cleanroom and air handling systems were much larger than the systems used in our lab, with the cleanroom space described as about 21,000 square feet. The facility also used ULPA filtration rather than only HEPA filtration, allowing smaller particles to be removed from the air. Another major difference was redundancy. The fan systems were designed with extra fans so that filtration could continue even if one component failed.

The level of defect control was also much stricter than in the lab cleanroom. Industrial wafer production separates spaces based on cleanliness and human access. For example, white rooms can be kept free of people, while grey rooms are used for maintenance access. This reduces human contamination and improves wafer quality. In our lab, we followed cleanroom procedures, but the process still involved much more manual handling and therefore more opportunity for contamination or process variation.

The field trip also showed that scaling semiconductor manufacturing is not only about making wafers larger. Although 450 mm wafers may seem like the next step, they have not been widely adopted because of cost. Instead, industry commonly uses 300 mm wafers because they provide a better balance between cost and throughput. The visit also showed the importance of advanced materials such as SOI wafers and silicon carbide, which are used for applications such as electric vehicles, aerospace, and high-performance electronics.

Photodiode Production Cost Analysis

A production cost estimate was completed using approximate IMSE tool rates, the process runsheet, the off-site ion implantation cost, and staff/TA labor. The calculation makes the assumption that one wafer contains 20 die and that each die contains about 25 photodiodes. Therefore, one wafer contains approximately 500 photodiodes.

The cost per die can be calculated as

$$\text{Cost per die} = \frac{\$2055}{20} \approx \$103 \text{ per die.}$$

The cost per photodiode can be calculated as

$$\text{Cost per photodiode} = \frac{\$2055}{20 \times 25} = \frac{\$2055}{500} \approx \$4.11 \text{ per photodiode.}$$

Table 5: Approximate production cost estimate for one silicon photodiode wafer.

Process or Material	Hours	Rate	Cost
Silicon wafer	–	\$15/item	\$15
Photoresist and chemicals	–	approx.	\$75
Oxidation furnace	10.0	\$3/hr	\$30
Ellipsometry	0.5	\$7/hr	\$4
Active lithography	0.75	approx.	\$12
Active oxide RIE etch	0.10	\$42/hr	\$4
Ion implantation	–	\$540/wafer	\$540
BOE/native oxide clean	0.25	\$10/hr	\$3
Annealing furnace	6.0	\$3/hr	\$18
Contact lithography	0.75	approx.	\$12
Contact opening RIE etch	0.15	\$42/hr	\$6
BOE clean before metal	0.17	\$10/hr	\$2
Aluminum sputtering	0.5	\$21/hr	\$11
Metal etch and photoresist strip	0.75	\$10/hr	\$8
Dicing saw	0.5	\$24/hr	\$12
Staff/TA labor	23.5	\$56/hr	\$1316
Total cost per wafer	–	–	\$2055

Based on this estimate, fabricating one wafer costs about \$2055. Since the wafer contains about 20 die, the cost is about \$103 per die. Since each die contains 25 photodiodes, the cost is about \$4.11 per photodiode. The two largest costs are staff/TA labor and off-site ion implantation. Staff and TA labor was calculated using the IMSE rate of \$56 per hour, while boron ion implantation was estimated at about \$540 per wafer from the outside processing quote.

This cost is much higher than the cost of a commercial photodiode because this process was done on a small scale in a teaching cleanroom. In industry, many wafers are processed at the same time with automated tools, so the cost of labor, setup, and tool time is spread across many more devices. In this lab, only a small number of wafers were processed, and many steps required manual handling, which increases the cost per device.

Envisioning Future Work

Future work should focus on repeating the fabrication process with better control of the photoresist and oxide etching steps. The hard bake, descum, and RIE time should be checked more carefully so that the oxide layer is not completely removed again. After the BEOL process is finished, the photodiodes should be tested using dark I–V, illuminated I–V, and responsivity measurements at different wavelengths.

Conclusion

This project demonstrated the design, fabrication, simulation, and planned characterization of a silicon pn photodiode. The work connected semiconductor device physics to real fabrication steps, including thermal oxidation, photolithography, reactive ion etching, ion implantation, annealing, and BEOL contact formation. The oxidation step successfully produced an oxide thickness close to the expected value, showing that the initial wafer preparation and furnace process were effective.

A major issue occurred during oxide etching, where the silicon dioxide layer was removed more than intended. This showed how sensitive semiconductor fabrication is to photoresist adhesion, hard baking, etch rate, and plasma exposure. Even one process error can prevent the final device from being completed properly.

Sentaurus simulations were updated to focus on the actual photodiode fabrication process. SProcess was used to simulate the final device structure and dopant formation, while SDevice was used to estimate the dark I–V behavior of the completed simulated device. The simulated dark current remained very low, while the measured devices showed more non-ideal behavior due to fabrication and contact issues. Overall, this project showed how fabrication process choices directly influence photodiode structure and performance, and future work should focus on improving etch control, completing BEOL processing, and performing full electrical and optical testing.

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